

RATNADEEP ABITKAR

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EDUCATION

Bachelor of Technology in Electronics and Telecommunication Engineering

KIT College of Engineering, Kolhapur

CGPA: 7.84

2021–2025

EXPERIENCE

App Developer

July 2026–Present

Mechatronics System Pvt Ltd

Pune, India

- Developed a production-ready **Flutter** mobile application for the company's **Real Time Data Acquisition System (RTDAS)** used for live dam monitoring, implementing a modular architecture with reusable widgets and a dedicated API service layer.
- Migrated the backend from **Node.js** to a **PHP REST API backed by Microsoft SQL Server**, rebuilding database queries and endpoints while maintaining compatibility with the existing dashboard functionality.
- Implemented secure authentication, live station monitoring, interactive maps using **flutter_map**, real-time telemetry dashboards, and **Syncfusion** gauges and charts with Historical, Daily, and Last 7 Days reporting.
- Integrated REST APIs, optimized JSON parsing and state management, resolved data inconsistencies between web and mobile dashboards, and generated Android release builds for internal deployment.

Full Stack Intern

March 2026–June 2026

iGap Technologies

Remote, India

- Built a full-stack **TV e-commerce platform** using the **MERN stack**, featuring a product catalog, shopping cart with discount code support, and a password-protected admin dashboard with full CRUD operations using MongoDB.
- Developed a **devotee management web application** for ISKCON Kolhapur with React.js and Express.js, implementing **JWT-based authentication**, admin user controls, a darshan gallery, Ekadashi calendar, and a devotional bookstore with live cart functionality.
- Progressed through the full stack independently—from HTML, CSS, JavaScript, and Bootstrap to React.js, Node.js, Express.js, and MongoDB—delivering two client-facing production projects.

PROJECTS

Purplexity

React 19, Express, Bun, TypeScript, Tavily API, Supabase

- Built an AI search engine combining the **Tavily Web Search API** and Google's **Gemini API** through the **Vercel AI SDK**, generating real-time streaming answers with dynamic web citations and URL attribution.
- Engineered multi-provider authentication with **Supabase OAuth** for Google and GitHub and implemented custom **Express.js** middleware to validate JWT bearer tokens and synchronize user sessions.
- Modeled relational schemas for multi-turn conversations and message histories using **Prisma ORM** with hosted **PostgreSQL** on Supabase, paired with a responsive **React 19** and **Tailwind CSS** frontend running on **Bun**.

Terminal-Based Coding Agent – Ongoing

TypeScript, Node.js, Gemini API, Claude API, ReAct

- Building an autonomous terminal coding assistant in **TypeScript**, inspired by Claude Code, with a multi-turn **ReAct (Reasoning + Action) loop** capable of iterative perception, autonomous tool chaining, and self-correction across up to 25 turns.
- Architected an extensible **dual-LLM provider engine** supporting **Google Gemini** via `@google/genai` and **Anthropic Claude** via `@anthropic-ai/sdk`, featuring live terminal streaming and a configurable thinking mode.
- Developing an autonomous tool suite featuring exact-match block editing through `edit_file` to prevent file truncation, recursive directory creation through `write_file`, and shell execution through `run_bash` in an interactive CLI REPL built with **Chalk** and **Ora**.

PaperCast – Ongoing

Next.js, TypeScript, Prisma, SQLite, Gemini, OpenAlex

- Building a full-stack application that converts research papers into interactive two-host AI podcasts, featuring a **Level-3 citation crawler** that recursively resolves direct citations and their seminal roots through the **OpenAlex API** into a structured knowledge graph.
- Engineering an LLM-driven script synthesizer using **Google's Gemini API** to generate multi-turn dialogue across the citation lineage, with a deterministic fallback generator to ensure episode creation without live API access.
- Designing an **interrupt-and-resume state machine** that allows users to pause playback through text or voice, receive a context-aware AI answer grounded in the citation graph, and resume from the exact paused segment.
- Modeling a relational schema in **Prisma** and **SQLite** for sessions, self-referencing citation hierarchies, podcast episodes, segments, and conversation history, with a three-pane NotebookLM-style interface built using Next.js, TypeScript, and Tailwind CSS.

TECHNICAL SKILLS

Languages

JavaScript, TypeScript, Dart, PHP, C, C++, Java

Frontend

React.js, Next.js, Flutter, Tailwind CSS, Radix UI

Backend

Node.js, Express.js, Bun, PHP, REST APIs

Databases

PostgreSQL, MongoDB, Microsoft SQL Server, Supabase

Frameworks & Libraries

Prisma, Vercel AI SDK, Git, Docker